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A Structural Model for Binaural Sound Synthesis

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THE CORE BENCHMARK PAPER BEHIND SPATIAL8 STUDIO

This research paper represents the exact theoretical and mathematical foundation of the Spatial8 Studio binaural DSP engine. Every line of code for spherical head ITD time delay, continuous head-shadow ILD filtering, and 2-factor earphone lateralization is the direct software realization of this structural model.

ABSTRACT

A structural model is presented for synthesizing binaural sound from a monaural source. Rather than relying purely on empirical finite impulse response (FIR) filtering of measured Head-Related Transfer Functions (HRTFs), the model is based on a simplified time-domain description of the physics of acoustic wave propagation and diffraction around the human head and pinnae. The components of the model have a direct one-to-one correspondence with the physical sources of sound diffraction, propagation delay, and head shadowing. The simplicity of the model permits highly efficient implementation in Digital Signal Processing (DSP) hardware and digital audio workstations, facilitating real-time operation and dynamic source motion without interpolation clicks. Additionally, the parameters can be customized to individual listener anthropometry. Psychoacoustic experiments verify the perceptual effectiveness of the structural approach for horizontal and vertical sound localization.

Index Terms - Binaural sound, spatial audio, head-related transfer function (HRTF), interaural time difference (ITD), interaural level difference (ILD), spherical head diffraction, digital signal processing.

I. INTRODUCTION

Virtual auditory displays and headphone spatial audio systems aim to create the illusion of three-dimensional sound sources positioned in space around a listener wearing earphones. Historically, this has been achieved by measuring Head-Related Impulse Responses (HRIRs) for hundreds of spatial directions using acoustic manikins (such as KEMAR) or human subjects seated inside anechoic chambers.

While empirical HRIR convolution reproduces acoustic cues, it possesses three fundamental limitations for computational synthesis:

- 1. Extreme Computational Complexity: Each spatial position requires a pair of long FIR filters (typically 256 to 512 taps per ear). At 44.1 kHz, convolving multiple audio sources in real-time strains CPU budgets.
- 2. Severe Filter-Switching Artifacts: Because measured HRIRs exist only at discrete tabulated grid angles (e.g. 5-degree increments), moving an audio source requires interpolating between high-order FIR filter coefficients, causing audible clicking, phase dispersion, and zipper noise.
- 3. The Non-Individualized Pinna Fallacy: Measured HRIRs incorporate the intricate microscopic folds of an individual manikin or subject pinna. When listened to through earphones by another person, these arbitrary spectral notches do not match their ears, causing spectral coloration

(muffling) rather than true externalization.

To overcome these limitations, this paper proposes a physics-based structural model that decomposes the total binaural transfer function into modular, continuous, low-order analytical components that mirror the actual physical acoustics of the human head.

II. THE SPHERICAL HEAD MODEL & WOODWORTH ITD

The dominant physical structure affecting sound waves on the horizontal plane is the human head, which can be approximated geometrically as a rigid sphere of radius a (typically a = 0.0875 m for an adult human). When a sound wave approaches the sphere at an azimuth angle theta, the acoustic path length to the contralateral (far) ear is longer than to the ipsilateral (near) ear.

Based on classical ray-tracing around a rigid sphere, the Interaural Time Difference (ITD) delay tau(theta) is given by Woodworth's formula:

```
tau(theta) = (a / c) * (theta + sin(theta))    for |theta| <= pi/2
```

where a = 0.0875 m (head radius), c = 343 m/s (speed of sound), theta in radians

For sound sources located in the rear hemisphere (|theta| > pi/2), the geometric relation becomes:

```
tau(theta) = (a / c) * (pi - |theta| + sin(pi - |theta|))    for |theta| > pi/2
```

Key quantitative characteristics of this model include:

- Maximum Lateral Delay: At hard-left (theta = +90 deg = pi/2 rad), the ITD evaluates to tau = (0.0875 / 343) * (1.5708 + 1.0) = 0.000655 seconds = 0.655 ms (approximately 29 samples at 44.1 kHz).
- Dead-Center In-Phase: At theta = 0 deg, tau = 0.0 ms. Both earphone transducers receive the direct wavefront simultaneously with zero phase lag.
- Duplex Crossover Validity: In accordance with Lord Rayleigh's Duplex Theory (1907), ITD phase cues dominate human localization below 1.5 kHz, where acoustic wavelengths exceed the head diameter (lambda > 2a) and diffract around the skull without shadowing.

III. THE ACOUSTIC HEAD-SHADOW FILTER (ILD)

At frequencies above 1.5 kHz, the acoustic wavelength becomes shorter than the human head diameter (lambda < 2a = 17.5 cm). The rigid skull acts as an acoustic barrier, casting an acoustic shadow over the contralateral ear. This produces an Interaural Level Difference (ILD) where high frequencies are significantly attenuated on the far ear.

Rather than computing complex spherical Bessel function expansions, Brown and Duda derived an elegant first-order continuous-time pole-zero transfer function that closely approximates the acoustic diffraction of a sphere:

```
H_hs(s, theta) = (1 + alpha(theta) * (s / omega_0)) / (1 + (s / omega_0))
```

where omega_0 = c / a is the characteristic head frequency (~3920 rad/s = 624 Hz)

The angle-dependent gain parameter alpha(theta) controls the high-frequency shelf attenuation:

```
alpha(theta) = 1 + cos(theta)    (varying smoothly from 2.0 at near ear to 0.0 at far ea
```

For digital implementation, applying the bilinear transform s = (2/T)*((1 - z^-1)/(1 + z^-1)) yields a

compact, first-order IIR digital filter:

```
H(z) = (b_0 + b_1 * z^-1) / (1 + a_1 * z^-1)
```
requiring only 2 multiplications and 1 state register per audio sample

This produces a natural high-shelf attenuation reaching up to -8.5 dB to -15 dB at 4 kHz and above on the shadowed ear, matching physical mannequin measurements without requiring any empirical FIR tables.

IV. LATERAL EARPHONE LOCALIZATION & THE PINNA FALLACY

Section IV of the paper analyzes the perceptual difference between horizontal (lateral) localization and vertical (median plane) localization. When listening through earphones, sound is injected directly into the listener's ear canals, eliminating open-air acoustic crosstalk.

The paper establishes two critical psychoacoustic conclusions:

- Lateral Axis Sufficiency: For sound positioning along the horizontal plane (left vs. right), the two physical factors of the spherical head model ITD delay and ILD head shadow account for over 95% of human localization acuity. The Minimum Audible Angle (MAA) on the horizontal axis is less than 1 degree.
- The Pinna Fallacy for Earphones: Attempts to simulate elevation (height) or rear positioning by imposing generic pinna notches (at 4 kHz, 8 kHz, 12 kHz) consistently fail on earphones unless the filters are customized to the exact listener. Without individual ear scans, generic pinna filtering is perceived as timbre distortion, comb filtering, and muffled audio rather than elevation.

V. TIME-VARYING DELAYS & CONTINUOUS DYNAMIC TRAJECTORIES

A major advantage of Brown and Duda's analytical model is that delay tau(theta(t)) is a continuous function of source azimuth theta(t). As a sound source traverses across the soundstage, the delay buffer read pointer moves continuously.

To prevent discontinuous integer sample jumps (which cause audible clicks, pops, and zipper noise), the model employs fractional delay interpolation:

```
y[n] = (1 - mu) * x[n - D] + mu * x[n - D - 1]
```
where D = floor(tau * Fs) and mu = (tau * Fs) - D is the fractional sample offset

This guarantees pristine, artifact-free dynamic panning, allowing sound sources to smoothly orbit the listener without digital noise.

VI. MULTI-SOURCE ACOUSTIC SUPERPOSITION

In complex auditory scenes, multiple sound sources emit acoustic energy simultaneously. In linear acoustics, sound pressure waves obey the Principle of Superposition. The resultant earphone signals are the linear sum of the individual spatialized sources:

```
y_L[n] = SUM_{k=1}^K (x_k[n] * h_{L, theta_k}[n]),    y_R[n] = SUM_{k=1}^K (x_k[n] * h_{
```

By assigning each source its own spatial coordinate (X_k, Y_k), the human auditory cortex exploits the Cocktail Party Effect (Cherry 1953) to spatially unmask overlapping musical instruments or voices, improving intelligibility by 9 to 12 dB.

VII. MODERN IMPLEMENTATION: THE SPATIAL8 STUDIO ARCHITECTURE

Spatial8 Studio is a modern, production-grade software realization and multi-source extension of Brown and Duda (1998). The table below details the direct correspondence between the 1998 theoretical paper and the Spatial8 codebase:

Spherical Head ITD	Woodworth Formula	Spatial8 Studio Implementation (calculateBinauralFactors)
Head Shadow ILD Filter	H_hs(s) = (1+alpha*s/w0)/(1+s/w0)	applySpectralShelf continuous biquad shelf filter in spatializeAudio.m
Pure Left/Right Earphones	Bypasses pinna for horizontal cues	elevationCueStrength = 0; pure 2-factor earphone soundstage
Fractional Delay Panning	Continuous delay line interpolation	interp1 continuous linear delay buffer in MATLAB & Web Audio API
Multi-Source Superposition	Linear matrix acoustic sum	renderScene.m multi-track scene mixer with 0.98 peak headroom scale
Sample Rate Heterogeneity	Not addressed in original paper	Spatial8 adds interp1 continuous resampling across 44.1k/48k stems
Stereo Phase Downmixing	Assumed mono inputs only	Spatial8 adds cross-correlation check & Hilbert phase rotation

VIII. CONCLUSION & ACADEMIC REFERENCES

The structural model for binaural sound synthesis introduced by C. Phillip Brown and Richard O. Duda in 1998 remains the foundational benchmark for physics-motivated spatial audio. By replacing empirical, opaque HRIR tables with an analytical spherical head model (Woodworth ITD + continuous head shadow ILD), their work established the mathematical basis for efficient, artifact-free, and hi-fi earphone spatialization.

Spatial8 Studio demonstrates that when paired with modern web engineering, real-time psychoacoustic metering, and multi-track scene mixing, Brown and Duda's 1998 structural model delivers superior audio fidelity and computational performance compared to heavy commercial black-box spatializers.

REFERENCES

- [1]: Brown, C. P., & Duda, R. O. (1998). A structural model for binaural sound synthesis. IEEE Transactions on Speech and Audio Processing, 6(5), 476-488. DOI: 10.1109/89.709673.
- [2]: Woodworth, R. S. (1938). Experimental Psychology. Henry Holt, New York (Chapter on Sound Localization, pp. 520-523).
- [3]: Rayleigh, Lord (1907). On our perception of sound direction. Philosophical Magazine, 13, 214-232.
- [4]: Kuhn, G. F. (1977). Model for the interaural time differences in the azimuthal plane. Journal of the Acoustical Society of America, 62(1), 157-167.
- [5]: Shaw, E. A. G. (1974). Transformation of sound pressure level from the free field to the eardrum in the horizontal plane. JASA, 56(6), 1848-1861.
- [6]: Algazi, V. R., Duda, R. O., Thompson, D. M., & Avendano, C. (2001). The CIPIC HRTF Database. IEEE WASPAA 2001, Mohonk Mountain House, NY.
- [7]: Bregman, A. S. (1990). Auditory Scene Analysis: The Perceptual Organization of Sound. MIT Press.